

Tensor Analysis

2. Spaces of N Dimensions

A point in three-dimensional space is a set of three numbers, called coordinates, determined by specifying a particular coordinate system or frame of reference. For example (x, y, z) , (ρ, ϕ, z) , (r, θ, ϕ) are coordinates of a point in rectangular, cylindrical, and spherical coordinate systems, respectively. A point in N -dimensional space is, by analogy, a set of N numbers denoted by $(x^1, x^2, \dots, x^N)$ where $1, 2, \dots, N$ are taken not as exponents but as superscripts, a policy which will prove useful.

The fact that we cannot visualize points in spaces of dimension higher than three has of course nothing whatsoever to do with their existence.

Coordinate Transformations

Let $(x^1, x^2, \dots, x^N)$ and $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N)$ be coordinates of a point in two different frames of reference. Suppose there exist N independent relations between the coordinates of the two systems having the form:

$$\begin{aligned}\bar{x}^1 &= \bar{x}^1(x^1, x^2, \dots, x^N) \\ \bar{x}^2 &= \bar{x}^2(x^1, x^2, \dots, x^N) \\ &\vdots \\ \bar{x}^N &= \bar{x}^N(x^1, x^2, \dots, x^N)\end{aligned}$$

which we can indicate briefly by

$$\bar{x}^k = \bar{x}^k(x^1, x^2, \dots, x^N) \quad k = 1, 2, \dots, N \dots\dots\dots(1)$$

where it is supposed that the functions involved are single-valued, continuous. and have continuous derivatives. Then, conversely to each set of coordinates $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N)$, there will correspond a unique set $(x^1, x^2, \dots, x^N)$ given by

$$x^k = x^k(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N) \quad k = 1, 2, \dots, N \dots\dots\dots(2)$$

The relations (1) and (2) define a transformation of coordinates from one frame of reference to another.

Summation Convention

Consider the expression $a_1x^1 + a_2x^2 + \dots + a_Nx^N$. This can be written using the short notation $\sum_{j=1}^N a_jx^j$. An even shorter notation is simply to write it as a_jx^j where we adopt the convention that whenever an index (subscript or superscript) is repeated in a given term, we are to sum over that index from 1 to N unless otherwise specified. This is called the summation convention. Clearly, instead of using the index j , we could have used another letter, say p , and the sum could be written a_px^p . Any index that is repeated in a given term, so that the summation convention applies, is called a dummy index or umbral index.

An index occurring only once in a given term is called a free index and can stand for any of the numbers $1, 2, \dots, N$ such as k in equation (1) or (2), each of which represents N equations.